

Makis Lam

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RESEARCH INTERESTS

◦ Dexterous Manipulation ◦ Robot Learning ◦ Tactile and Proprioceptive Sensing ◦ Soft Robotics ◦ Robot Control

EDUCATION

University of Waterloo

Sept. 2022 – Apr. 2027

Bachelor of Applied Science, Honours Mechanical Engineering, Co-op (3.81/4.0 GPA)

Waterloo, ON

PUBLICATIONS

- **M. Lam**, C. Bauer, D. Bauer, and N. Pollard, “Learned Remote Proprioception and Contact Force Estimation from Motor and Load Cell Signals,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2027. Under review.

RESEARCH EXPERIENCE

FuturHand Robotics, in collaboration with Carnegie Mellon University

May 2026 – Present

Robotics & Research Engineer | Prof. Nancy Pollard, Dr. Dominik Bauer, Dr. Cornelia Bauer

Pittsburgh, PA

- Developed learned models (GRU/MLP) that estimate contact onset, contact force, and joint angles from motor and load cell signals, achieving 4.11° joint-angle MAE, 0.34 N contact-force MAE, and 0.79 F1 for contact onset.
- Deployed real-time inference at 90 Hz on a tendon-driven hand; first-author paper under review at ICRA 2027.
- Designed multiple underactuated, tendon-driven soft robotic hands, using motion capture data to optimize joint kinematics for tendon force transmission, and FDM-printed them in flexible materials.
- Developing an automated CAD pipeline that generates hands from optimized kinematics for task-specific designs.

University of Waterloo SMART-Lab

Sept. 2025 – Present

Robotics Undergraduate Research Assistant | Prof. Hamed Shamsavan

Waterloo, ON

- Designed a rotating end-effector for a 6-axis arm that generates dynamic magnetic fields to actuate microrobots.
- Developed an OpenCV script to measure microrobot displacement across 30+ experiments in varying conditions.

University of Waterloo Composites Research Group

May – Aug. 2024

Undergraduate Research Assistant | Prof. John Montesano

Waterloo, ON

- Designed and fabricated punch geometries to produce specific deformation modes in unidirectional carbon fabric.
- Measured fabric deformation with digital image correlation (DIC) to validate a fabric-forming simulation model.

INDUSTRY EXPERIENCE

Formlabs

Jan. – Apr. 2026

Mechanical Design Engineer | SLS R&D

Somerville, MA

- Owned end-to-end design of the Fuse X1 air-handling subsystem and co-developed optical subsystems, delivering production CAD and driving DFMA to achieve 95%+ part yields.
- Designed, built, and programmed a 50+ part laser purge characterization jig, and deployed it in manufacturing.
- Modeled optical purge airflow in Ansys Fluent using porous media models to select materials for high-temperature environments, while optimizing for flow uniformity; validated the simulations against in-printer measurements.
- Built a compression molding press with 3D-printed tooling and PID temperature control for composite fabrication.

printbnb

May 2023 – Dec. 2025

Co-Founder & CEO

Waterloo, ON

- Co-founded a web-based marketplace that connects a network of decentralized private 3D printer operators with customers, as an alternative to commercial 3D printing services.
- Grew the platform to 8+ printer operators and 150+ unique fulfilled orders for individual and business customers, facilitating 20+ printed parts and custom designs per month; selected for the Velocity Cornerstone incubator.

Tesla

May – Aug. 2025

Robotics Hardware Engineer | Optimus

Palo Alto, CA

- Analyzed end-effector error sources from first principles and designed dimensional measurement strategies, identifying key factors that limit robot accuracy and controllability.
- Developed motion capture (MOCAP) metrology procedures to validate calibration repeatability and accuracy; designed experiments (DOE) and analyzed results statistically, reducing assembly time by 15 minutes per robot.
- Analyzed mainline operations using PFMEA to improve operator ergonomics and balance station workloads, reducing cycle time, increasing unit output, and supporting the Optimus line's transition to mass production.

Formlabs

Sept. – Dec. 2024

Settings Optimization Engineer | SLA Printers

Somerville, MA

- Investigated flowmark surface artifacts in SLA prints, identifying root causes and the conditions that make them visible; developed a Python script that modifies the Form 4 print process to eliminate the defects.
- Tuned 6+ print resins via first principles, owning ~3% of Form 4L settings and informing printer design reviews.
- Increased LCD panel lifetime testing efficiency by 57% by proposing and validating tests at smaller layer heights.

Airbus

Jan. – Apr. 2024

Integration Engineer | A220 Cabin

Mirabel, QC

- Conducted feasibility studies and developed design solutions for 15+ customer requests, presenting results to airline customers and influencing the final cabin layout and monument configuration of their A220 aircraft.
- Automated drawing tracking in Google Apps Script, helping cut head-of-version lead time from 12 to 9 months.

Delta Elevator Co. Ltd.

May – Aug. 2023

Mechanical Design Engineer

Kitchener, ON

- Designed a set of 3 adjustable oil buffer stands, validated with Ansys FEA, removing the need for custom stands.
- Redesigned cable guard joints with DFMA, reducing installation hardware and extending component life by 25%.

TECHNICAL SKILLS

- **Programming:** Python, C++, MATLAB, PyTorch, OpenCV
- **Robotics:** ROS 2, MoveIt 2, Sensor Fusion, State Estimation, Motion Capture
- **Machine Learning:** Deep Learning, Time-Series Modeling, Imitation & Reinforcement Learning, VLA Models
- **CAD & Simulation:** Onshape, SOLIDWORKS, CATIA V5/V6, AutoCAD, KiCad (PCB), Ansys FEA & Fluent
- **Fabrication & Testing:** 3D Printing (SLA/SLS/FDM), DIC, DOE, GD&T, DFMA, DFMEA

AWARDS & HONOURS

- **President's Research Award**, University of Waterloo 2026
- **Best Use of Bracket Bot Hardware**, Hack the North (Canada's Largest Hackathon) 2026
- **President's International Experience Award**, University of Waterloo 2025
- **Term Distinction (Dean's List)**, University of Waterloo Winter & Fall 2025
- **President's Scholarship of Distinction**, University of Waterloo 2023
- **CAD Designathon Winner**, McMaster Design League 2023
- **Air Line Pilots Association of Canada Award**, Air Line Pilots Association 2022
- **Canada Company Youth Excellence Award**, Canada Company 2022
- **Lord Strathcona Trust Fund Medal**, Lord Strathcona Trust 2022

LEADERSHIP & VOLUNTEERING

- **UW Chinese Christian Fellowship**, Chair, Vice Chair, Executive Member 2024 – Present
- **Mecha Mayhem** (Canada's Largest Robotics Competition), Key Volunteer & AV Coordinator 2023 – Present
- **KW Redeemer Presbyterian Church**, AV Volunteer 2023 – Present
- **UW Varsity Ultimate Frisbee**, Team Captain (2023) 2022 – 2025
- **Waterloo Rocketry**, Curing Oven Lead 2022 – 2023
- **351 Silver Star Royal Canadian Air Cadets**, Warrant Officer Second Class 2017 – 2022

CERTIFICATIONS

- **Private Pilot Licence (PPL)**, Transport Canada 2022
- **Basic Drone Operator Licence**, Transport Canada 2020